Land at Sladburys Lane, Little Clacton

Ecological Impact Assessment

Client: Stanfords

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Ref: 1930

Issue: Rev A

Pond House

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[redacted]

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1.0 SUMMARY

1.1 The site (located at central grid ref: TM 19061 16815) was found to comprise an open area of other neutral grassland and a significant area of compacted hard core used for parking and storag of materials. The field is surrounded on three sides to varying degrees by mature hedges with trees. Outline planning permission is being sought to provide 29 residential units on the site, accessed via Sladbury's Lane to the south east.

1.2 The trees, hedges and to a lesser degree the grassland provide suitable habitat for nesting birds. Ideally, works will commence during September to February inclusive to avoid the bird nesting season. If this is not possible, immediately prior to commencement of works a check for nesting birds should be undertaken by a suitably experienced ecologist. Any active nests will need to be left in situ until the young have left the nest.

1.3 There is low potential for occasional transient reptiles to be present in small areas of long ruderal vegetation / grass between the grass field and the parking area / compoun commencement of works, this vegetation should be subject to a precautionary staged clearance using hand held strimmers. The grass should be strimmed to 150mm height, and left for 2 hours, before being strimmed to ground level.

1.4 2025 eDNA surveys of two ponds located within 250m of the site returned a negative result for the presence of great crested newt DNA, and this species are unlikely to be present on adversely affected by the proposals.

1.5 The proposals are unlikely to result in any significant adverse impacts upon any other protected or notable species or habitats.

1.6 The mitigation measures detailed in sections 5.0 and 6.0 can be secured via a planning condition, and could result in a minor overall enhancement at the site level for a number of suburban bird species, roosting bats and amphibians.

2.0 INTRODUCTION

Instruction

2.1 This report has been prepared by Liz Lord following instruction by Ms. A Cox of Stanfords to carry out an ecological appraisal of land to the north west of Sladbury's Lane, Little Clacton, Essex.

Site Proposals

2.2 Outline planning permission is being sought to provide 29 residential units on the site, accessed via Sladbury's Lane to the south east.

Site Description

2.3 The site lies on the eastern outskirts of Little Clacton, and to the north west of Holland-on-Sea, near Clacton, Essex. It is adjoined to the south and west by existing and new residential dwellings. To the north and east is arable farmland, extending north to the edge of Little Clacton. Sladbury Lane runs along the south east site boundary.

2.1 The wider landscape is dominated by residential dwellings, interspersed with small pockets of grassland / scrub parkland, commercial buildings and caravan parks, with arable fields extending north east. Grazing pastures associated with Holland Brook are present c.1km to the east. Habitats within the surrounding landscape are moderately well connected via hedgerows and lines of trees.

Fig 1: Survey / site boundary highlighted red.

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Objectives

2.2 This report has been written broadly in accordance with the report writing guidelines produced by the Chartered Institute of Ecology and Environmental Management (CIEEM) (CIEEM 2018, 2017a, 2017b). In accordance with the client brief, this survey and report aims to:

2.2.1 Where possible, identify and describe all potentially significant ecological effects on protected and notable species / sites associated with the proposals;

2.2.2 Where possible, set out the mitigation measures required to ensure compliance with nature conservation legislation and address any potentially significant ecological effects;

2.2.3 Identify how mitigation measures will / could be secured;

2.2.4 Provide an assessment of the significance of any residual effects;

2.2.5 Identify appropriate enhancement measures; and

2.2.6 Where deemed necessary, set out the requirements for post construction monitoring.

2.3 This survey and report is intended to inform, as necessary, the layout and design of the proposals, future landscape design and management on site, and where required the methodology and timing of development works.

Timescales

2.4 The construction period is expected to be around 18-24 months following the granting of relevant permissions.

2.5 This report is valid for a period of 12 months from the date of survey. Beyond this time, changes to the vegetation may have occurred which could require re-assessment and potentially further survey to re-determine the presence / likely absence of protected species.

Relevant Documents

2.6 The site assessment was based upon drawing number 3735 PA-10 dated May 2025 by Duncan Clark and Beckett Ltd, as shown in Appendix 1. Note that any minor amendments to the overall scheme are unlikely to alter the conclusions and recommendations of this report.

2.7 Recommendations included within this report are the professional opinion of an experien ecologist based on the client's proposals for the site, the site surveys, the results of the desk study, and features present in the surrounding environment.

3.0 METHODOLOGY

Desk Study

3.1 The Multi Agency Geographic Information for the Countryside (MAGIC) we on 24th April 2025 to determine the presence of any nationally and internationally designated site such as Sites of Special Scientific Interest (SSSI), Special Areas of Conservation (SACs), Special Protection Areas (SPAs) and Ramsar sites within influencing distance of the proposals.

3.2 The MAGIC website was also used to search for any records of European Protected Spe Mitigation (EPSM) licences that have been approved by Natural England within a 5km radius c the application site since late 2008 (last updated May 2024). The website was checked for any data from Natural England's great crested newt eDNA Habitat Suitability Index pond surveys fo District Level Licensing 2017-2019 (last updated June 2024); and data from Natural England great crested newt Class Survey Licence returns within a 5km radius of the site (last updated June 2024).

3.3 A records search was carried out in January 2025 with Essex Field Club for records of protected and notable species within 2-3km of the site central grid reference. The full results of the data search are available on request.

Site Survey

3.4 A daytime site survey was carried out on 23rd September 2024. The survey was based upon the standard methodology for Extended Phase 1 Habitat Surveys (JNCC 2010) and the UK Habita Classification system (UKHab Ltd 2023). The relative abundance of individual plant species was recorded, and habitats were classified according to the abundance of plant species pres Any evidence of invasive species such as Japanese knotweed was noted.

3.5 The relative abundance of individual plant species was recorded as per the requirements of the relevant condition assessments for the various habitats present on site (Defra, 2024), and habitats were classified according to the abundance of plant species present. Any evidence of invasive species such as Japanese knotweed was noted.

3.6 The survey encompassed all land within the red line boundary as shown in Figure 1, plus land immediately adjacent to the site, where accessible or visible.

3.7 References in this report and the accompanying Statutory Biodiversity Metric to 'domi 'abundant', 'frequent', 'occasional' and 'rare' link directly to the DAFOR scale for meas vegetation cover, as is referred to in the UKHab classifications, with the corresponding percentage area cover detailed in the table overleaf.

DAFOR term	Species percentage abundance (cover)
Dominant	76-100%
Abundant	51-75%
Frequent	26-50%
Occasional	1-25%
Rare	<1%

3.8 The survey also included an assessment of the site's potential to support any legally protected species; or Species and Habitats of Principal Importance, as identified by Section 41 of the Natural Environment and Rural Communities Act 2006. Where best practice guidelines exist, these have been used to assess the likelihood that individual species will be present, for example Bat Surveys: Good Practice Guidelines (Collins, J. 2023) and Habitat Suitability Index for Great Crested Newt (Oldham *et al*, 2000).

3.9 Using criteria provided in best practice guidelines, habitats have been assessed for their potential to support protected species; notably bats, barn owls *Tyto alba,* badgers *Meles meles*; great crested newts *Triturus cristatus*; reptiles, water voles *Arvicola amphibius*; dormice *Muscardinus avellanarius* and otters *Lutra lutra*.

3.10 Where methodologies, classification or recommendations deviate from best practice guidelines this report provides ecological justification for such changes.

Tree Survey

3.11 Trees were surveyed from ground level with a powerful torch and a pair of Nikon 12 x 50 binoculars for potential roost features (e.g. splits, cavities, lifted bark, woodpecker holes), and assessed in accordance with criteria outlined in Bat Surveys for Professional Ecologists: Good Pra Guidelines (Collins, J. 2023).

Habitat Suitability Index (HSI) assessment

3.12 Where relevant, for each water body located within potential influencing distance construction zone boundary (250m in this case), a Habitat Suitability Index (HSI) assessment was undertaken, following standard methods described in Oldham R.S. *et al*, (2000).

3.13 Features such as shading, water quality, terrestrial habitat, fish and fowl presence were not during the survey. These features were used in the HSI to assess the potential of the ponds support great crested newts.

3.14 Following the survey, the HSI field scores are inserted into a table to calculate a score for eac pond, with pond suitability for great crested newts assessed on the following scale:

HSI Score	Pond Suitability
< 0.5	Poor
0.5 – 0.59	Below Average
0.6 – 0.69	Average
0.7 – 0.79	Good
>0.8	Excellent

Great Crested Newt eDNA testing

3.15 Two water bodies were subject to an environmental DNA (eDNA) test for great crested n presence / absence. The kits were supplied by SureScreen Scientifics and the water samples were taken on 17th April 2025 by Liz Lord.

3.16 Liz is a licensed GCN surveyor and attended an ADAS-led training course on the methodology and application of eDNA testing in April 2017. The samples were collected, stored and couriered in accordance with Natural England protocol (Biggs *et. al* 2014), and were subject to a quantitative Polymerase Chain Reaction (qPCR) test. The full results can be found in Appendix 3.

3.17 The samples collected were of acceptable quality, and no limitations were highlig Surescreen Scientifics during the testing process.

Surveyors

3.18 The site survey was carried out by Liz Lord. Liz has been a professional ecologist since 2005, and holds current Natural England licences to survey bats - Class Licence Reg. No. 2015-13305-CLS-CLS; great crested newts - Class Licence Reg. No. 2020-44816-CLS-CLS; and barn owls - Class Licence Reg. No. CL29/00160. Liz is a full member of CIEEM.

3.19 The weather at the time of the survey was overcast with a light breeze (BF2-3) and a temperature of 16˚C.

Zone of Influence

3.20 The potential impacts of a development are not always limited to the boundaries of the sit concerned, such as where there are ecological or hydrological links beyond the site boundaries. In order for the proposed works to have an impact on habitats and species outside of the : boundaries, there needs to be a source of impact, a pathway and a receptor for that impact.

3.21 The Zone of Influence will vary for different habitats and species depending on their sensitivity to predicted impacts, the distribution and status of the relevant species, whether a species is mobile, migratory, and whether its presence and activity varies according to the seasons.

3.22 An assessment of the Zone of Influence has been made based on the site boundaries shown in Figure 1, and where necessary recommendations to avoid any significant adverse im beyond the site boundaries have been provided in section 5.0.

Limitations

3.23 The conclusions in this report are based on the best information available during the report period of survey.

3.24 Ecological surveys provide only a 'snapshot' of the site in time, and many species, such as bats and badgers, are capable of colonising a site in a very short space of time. Lack of evidence of a species at the time of survey can only allow conclusion of the *likely* absence of this species, since no level of survey effort is capable of proving absence beyond doubt.

3.25 Whilst best efforts have been made to identify all water bodies within 250m of the site, it is r always possible to record all garden ponds using Ordnance Survey maps and aerial photography. Additional search effort with respect to garden ponds is likely to be disproportionate, as many garden ponds have limited suitability for great crested newts, and it is a common cons associated with all Ecological Assessments.

3.26 The grassland appeared to have been mown within the last 4-8 weeks, with accumulated cuttings in some places. Whilst the shorter vegetation made the UKHabs survey more difficult, there was sufficient regrowth and low level vegetation remaining to be able to record and identify a significant number of species. Given that historical Google Earth imagery shows the site was ir arable production until around 2019/20, and that the species recorded reflect that of a typ floristically enhanced arable grass margin, the categorisation of the grassland as 'other neutral grassland' in 'moderate' condition is deemed to be reasonable and most likely accurate. T grass cutting is therefore not considered to be a significant constraint to the conclusions recommendations of this report.

Geographic Context

3.27 Where applicable, the importance of each ecological feature has been consid geographic context as follows:

International and European

National

Regional

Metropolitan, County, vice-county or other local authority-wide area

River Basin District

Estuarine system/Coastal cell

Local (further categorized into District, Borough or Parish)

Site

Assessment of Impacts and Effects

3.28 The following definitions are used for the terms ‘impact’ and ‘effect’ in accordance with CIEE (2018) guidelines:

Impact – actions resulting in changes to an ecological feature

Effect – outcome to an ecological feature from an impact

3.29 The importance of any ecological feature has been determined via the site surveys detailed in this report. Note that species and habitats afforded legal protection are, by default, alv considered within the EcIA assessment process to be ‘important’.

3.30 Potential impacts of the proposals on any such features have been assessed based on the client proposals for the site, and following a review of all phases of the project. Impacts are assessed through consideration of the extent, magnitude, duration, reversibility, timing and frequency of works which may result in likely ‘significant’ impacts to any ecological features present. The route through which impacts may occur (direct, indirect, secondary or cumulative) has also been considered. Positive impacts are assessed as well as negative ones.

3.31 The results of the surveys have been used to identify any potentially significant impacts in t absence of any avoidance, mitigation or compensation measures. Any such appr measures have then been proposed where necessary.

Characterisation of Ecological Impacts

3.32 When considering ecological impacts and effects, the following characteristics hav considered:

positive or negative
extent
magnitude
duration
frequency and timing
reversibility

3.33 Where various characteristics have not been specifically referred to in this report, they have been considered insignificant or irrelevant to that specific feature.

3.34 A ‘significant effect’ is defined within the current CIEEM guidelines (2018) as: “ *an effect that either supports or undermines biodiversity conservation objectives for ‘important ecological features’ or for biodiversity in general. Conservation objectives may be specific (e.g. for a designated site) or broad (e.g. national/local nature conservation policy) or more wide-ranging (enhancement of biodiversity). Effects can be considered significant at a wide range of scales from international to local.”*

3.35 Where a significant effect is predicted, this requires assessment and reporting in order to provide the decision maker with sufficient information to determine the environmental consequences of a project. A significant effect can be either positive or negative, and its extent will determine the requirement of conditions, restrictions or monitoring works.

3.36 The current CIEEM guidelines (2018) also state that: *“After assessing the impacts of the proposal, all attempts should be made to avoid and mitigate ecological impacts. Once measures to avoi and mitigate ecological impacts have been finalised, assessment of the residual impacts shoulc be undertaken to determine the significance of their effects on ecological features. Any residual impacts that will result in effects that are significant, and the proposed compensatory measures will be the factors considered against ecological objectives (legislation and policy) in determining the outcome of the application.”*

3.37 This report has taken into account the factors detailed above for each important ecol feature in the absence of mitigation. Recommendations have then been made with respect avoidance / mitigation / compensation / enhancement as necessary, and an assessment of the residual impacts after such measures has been made.

Mitigation Hierarchy

3.38 In order to minimise the likelihood of any significant negative residual effects on environment features, this assessment has followed the mitigation hierarchy (listed below in order of preference):

Avoidance – measures that avoid harm to ecological features, both spatially and temporally;
Mitigation – avoidance or minimisation of negative effects through appropriate timing of works, or the provision of mitigation measures within the scheme design which can be guaranteed by condition or similar;
Compensation – measures taken to offset residual effects which result in the loss of, or permanent damage to, ecological features despite mitigation;
Enhancement – measures to provide net benefits for biodiversity, either by improved management of existing features, or the provision of new features, and over and above that which is required to mitigate / compensate for an impact. Delivery should be secured via planning condition or similar.

Legislation and Policy

3.39 Specific reference has been made to the individual legal protection of the species detailed within this report, however additional information with respect to other relevant legislation and planning policy is provided in section 8.0.

3.40 The legislation of particular relevance within the body of this report is the Conservation of Habitats and Species Regulations 2017 (as amended) and the Wildlife and Countryside Act 198 amended). The former confers legal protection to 'European' Protected Species against both disturbance and harm, and extends to the full protection of their habitats. This legislation also provides legal protection for a number of internationally designated sites within the UK, remains in place following Brexit.

3.41 The Wildlife and Countryside Act 1981 (as amended) is UK specific, and generally only provides protection against direct harm to individuals of a species.

4.0 RESULTS (*Baseline Conditions*)

Site Summary

4.1 The site comprises an open area of other neutral grassland and a significant area of compacted hard core used for parking and storage of materials.

Desk Study: Statutory Designated Sites

4.2 The MAGIC website indicates that there are two statutory designated sites of national importance within 2km of the proposed development – Holland on Sea Cliff SSSI, located c.2km to the east of the site; and Holland Haven Marshes SSSI, located c.1km to the north east of the site.

4.3 There are no national or international statutory designated sites present within direct influencing distance of the site; however the site lies within the Zone of Influence for internationally designated Essex coastal sites due to the draw of the coast for recreation. This includes Hamford Water SAC, SPA and Ramsar site, located c.6.6km north east of the site; and the Colne Estuary (Mid Essex Coast Phase 2) SPA and Ramsar site, and the Essex Estuaries SAC, which both lie around 7.5km to the west.

Desk Study: Non-Statutory Designated Sites

4.4 There are no Local Wildlife Sites located within influencing distance of the site. Te100 Bursville Park and Te105 Bursville Park Cemetery are located just over 300m to the north east, separated from the site by an arable field. Both are designated for the grassland habitats present, as well as the value of the sites as urban green space, and have no direct connectivity to the pr development site.

Habitats

Invasive species

4.5 No aerial evidence of Japanese knotweed *Fallopia japonica* was recorded within the site or the immediately adjacent areas at the time of survey.

Other neutral grassland (g3c)

4.6 The vast majority of the site comprises 'other neutral grassland', which historical aer photographs and species composition suggest was sown with an agricultural floristicall enhanced mix around 2019/20. Despite the site showing evidence of supporting what was originally a floristically enhanced arable field, the habitats have not been classified as 'cropland' habitats under the UKHab classification system, as the site is no longer in an arable rotation, and does not appear to have been since around 2019/20.

4.7 The grass was relatively short at the time of survey, having been mown w -8 weeks, however significant regrowth was present as well as remaining low growing vegetation. Grass cover dominated at c.70% cover, comprising a relatively even mix of red fescue *Festuca rubra* (O-F), perennial ryegrass *Lolium perenne* (O-F), false oat grass *Arrhenatherum elatius* (F), smaller cats tail *Phleum bertolonii* (O-F), creeping bent *Agrostis stolonifera* (O-F), crested dogs tail *Cynosurus cristatus* (O) and Yorkshire fog *Holcus lanatus* (O). Variable moss cover was present at ground level.

4.8 Forbs averaged around 30% cover, ranging from 15-40%. The species recorded are detailed in Table 1, below. The number of species per m² was generally between five and eight.

Table 1: Number and frequency of forbs recorded across grassland

Species	DAFOR Frequency	Notes
Lesser knapweed *Centaurea nigra*	***Frequent***	Frequent throughout
Creeping thistle *Cirsium arvense*	***Occasional to frequent***	Present throughout, local abundance in places
Ribwort plantain *Plantago lanceolata*	***Occasional***	
Red clover *Trifolium pratense*		Locally abundant
Common ragwort *Senecio jacobaea*		
Birds foot trefoil *Lotus corniculatus*		
Creeping buttercup *Ranunculus repens*	***Rare***	
Sow thistle *Sonchus oleraceus*		
Groundsel *Senecio vulgaris*		
Scentless mayweed *Tripleurospermum inoc*		
Ox-eye daisy *Leucanthemum vulgare*		
Hedge bedstraw *Galium mollugo*		
Common hogweed *Heracleum sphondylium*		
Rosebay willowherb *Chamaenerion angustifolium*		

Native hedgerow with trees and ditch (h2a, 11, 50)

4.9 The north western site boundary is marked by an unmanaged hedgerow with trees, growing out to c.4m wide in places. Shrub and climbing species comprise blackthorn *Prunus spinosa*, holly *Ilex aquifolium*, elm *Ulmus sp.*, dog rose *Rosa canina* and bramble *Rubus fruticosus agg*. Mature trees are generally oak *Quercus robur* with occasional field maple *Acer campestre.* Most of the oaks were found to have minor deadwood and occasional branch splits, but were not viewed f the northern side. The shrubs and trees grow over a moderately deep ditch, dry at the western end and running wet (<10mm deep) at the eastern end where agricultural drainag discharge into the ditch. Where tree / shrub cover is not dense, ground flora comprises grasses, bracken *Pteridium aquilinum*, creeping thistle, hedge bindweed *Calystegia sepium* and nettle.

4.10 The far western end of the hedgerow does not support any tree or typical hedgerow shrubs, and comprises a small stand of dense bramble.

Native hedgerow with ditch (h2a, 50)

4.11 The western two thirds of the north eastern boundary continues this hedge, but with no trees. Ditch banks are covered in dense bramble, bracken and bindweed at the eastern (roadside) end.

4.12 The south eastern hedgerow is also unmanaged, and is dominated by blackthorn with some rose, bramble and occasional elm. The hedge grows up to c.5m wide in places and grows over a narrow (<200mm wide) ditch holding less than 15mm of water. A variable understorey of bracken, low brambles, nettle and hogweed *Heracleum sphondylium* is present beneath and immediately adjacent to the hedge.

Trees

4.13 A small number of individual trees grow immediately offsite along the south western site boundary – a small oak in the south western corner, an oak sapling and a mature oak pollard. The pollard was noted to have a small number of crevices and cavities, but little deadwood. Also pres between the trees are individual semi-mature hawthorn *Crataegus monogyna*, dog rose and a young goat willow *Salix caprea*.

4.14 A mature, multi-stemmed field maple tree grows onsite along the south eastern boundary, on the ditch banks.

Developed land; sealed surface (u1b)

4.15 The site is accessed via a tarmac road, which leads into the new housing estate offsite to the west. Direct access into the estate was restricted at the time of survey, with the road predominan being used to access a large parking area on the south eastern side of the site. The grassland and north eastern hedge is separated from the road and parking areas by continuous he fencing, or small lengths of hoarding.

Artificial unvegetated, unsealed surface (u1c)

4.16 The south eastern corner of the site consists of compacted hardcore and earth, used predominantly to park vehicles but in small areas to store skips and building materials on pallets.

Sparsely vegetated urban land – ruderal / ephemeral (u1f, 81)

4.17 A large mound of earth and small pieces of hardcore approximately 3-4m tall is present close to the western site boundary, likely associated with the construction of housing immediately offsite to the west. The mound has become partially colonised with ruderal / ephemeral species dominated by fat hen *Chenopodium album,* with some sow thistle, fumitory *Fumaria officinalis*, bristly oxtongue *Helminthotheca echioides*, musk mallow *Malva moschata*, yarrow *Achillea millefolium* and black nightshade *Solanum nigrum*.

Tall forbs (g3, 16)

4.18 Along the eastern boundary, close to the current site entrance, is an area of variable vegetation cover resulting from disturbance during laying of parking materials and herasfencing. Vegetation comprises occasional grasses such as false oat grass, with frequent creeping thistle, knapweed, bristly oxtongue, mugwort *Artemisia vulgaris* and small areas of bracken, bramble and blackthorn growing along the road boundary.

Site Photographs

Photo 1: Road into site, with grassland to north an parking area to south

Photo 2: Compacted hardcore and earth parking area

Photo 3: Stored materials and earth / rubble m close to western boundary

Photo 4: Mature offsite pollard oak tree along wester boundary / western side of grassland

Photo 5: Central area of grassland, looking north w

Photo 6: Eastern side of grassland, looking north ea

Photo 7: Grassland viewed from west to east

Photo 8: Typical view of grassland, with frequent larg clumps of knapweed, thistle regrowth, ribwort plantain and grass regrowth

Photo 9: Western boundary

Photo 10: North western boundary hedge with tre

Photo 11: North eastern boundary hedge (no trees

Photo 12: South eastern road verge with ruderal vegetation and mature multi-stemmed fiel maple

Water bodies

4.19 No water bodies are present on site. Ordnance Survey maps at 1:10,000 scale and photographs highlighted the presence of two water bodies within 250m of the site – WB1, an infield pond located c.210m to the north of the site; and WB2, a new likely balancing pond loc c.125m to the south of the site. Historical aerial imagery suggests WB2 was constructed around 2022/23. Following an HSI assessment, WB1 was found to be of 'good' suitability for great crested newts (GCN) (score 0.7) and WB2 was found to be of 'excellent' suitability for GCN (score 0.85).

Amphibians

4.20 The MAGIC search did not highlight any records of great crested newt (GCN) within 5km of the site, with a single confirmed negative survey at c.3.5km north east. The Essex Field Club records search did not return any records of GCN within 2-3km of the site.

4.21 The boundary hedges provide small areas of high quality potential habitat for amphibians, whilst the grassland provides low to moderate quality potential habitat depending upon the length the vegetation. The pile of earth and rubble is generally well consolidated, with few accessible crevices for amphibians, and the parking areas and access road are not suitable for amphibians.

4.22 WB1 and WB2 were subject to updated HSI assessments and - as a direct result - eDNA surveys in April 2025 to determine the presence / likely absence of GCN. The results showed that GCN are likely to be absent from both water bodies despite the ponds being of 'good' and 'excellent' suitability for GCN. GCN are therefore very unlikely to be present on site, and the potential for GCN to be adversely affected by the proposals is considered to be negligible.

<u>Bats</u>

4.23 The desk study identified one bat EPSM licence within 5km of the site, at 1.5km north west for a breeding roost of common pipistrelle *Pipistrellus pipistrellus* and soprano pipistrelle *P. pygmaeus.*

4.24 The records search returned a further 12 records of bats within 2-3km of the site, of common pipistrelle, unidentified pipistrelle and brown long-eared bat *Plecotus auritus*, including a brown long-eared maternity roost. Aside from one common pipistrelle and one brown long-eared bat record at c.0.5km to the south east of the site, all of the records were around 1.5km away.

Bats - roosting

4.25 No buildings are present on site, and no buildings with potential to support roosting bats we noted within potential influencing distance of the site.

4.26 The mature pollard oak tree located just outside the south western boundary was assessed as being of 'moderate' suitability for roosting bats, due to the presence of a small number of cavities. The mature oak trees present along the north western boundary are of 'low' suitability, with the field maples also of 'low' suitability. The multi-stemmed field maple growing on the south eastern boundary was assessed as being of negligible suitability, with no potential roosting features noted. However, it is noted that all but one of these trees (the multi-stemmed field maple) will be retained as part of the proposals.

Bats – commuting / foraging

4.27 The boundary hedgerows and to a much lesser degree the small number of trees along the south western site boundary provide moderate quality linear features with potential to be used by a range of foraging or commuting bats. Due to variable quality habitat connectivity beyond the site to areas of higher quality foraging habitat or to potential roosing features, the hedges are likely to be of limited importance to foraging / commuting bats, however all are due to b retained.

Reptiles

4.28 At the time of survey the vast majority of the site did not provide suitable habitat for reptiles, and the site is not currently connected to any areas of offsite potential reptile habitats. However the data search returned one record of common lizard *Zootoca vivipara* and slow worm *Anguis fragilis* from within the new housing development offsite to the south west, dating from 2016, indicating the past recent presence of both species here. These records appear to be associated with a former field boundary. Records of adder *Vipera berus* and grass snake *Natrix helvetica* were also returned from the wider landscape, over 1.5km from the site.

4.29 Longer margins of vegetation – predominantly ruderal vegetation, with some long grass – are present along the south eastern boundary between the parking areas and Sladbury's Lane, and around the base of the spoil pile. Given the past local presence of both slow worm and common lizard, there is moderate potential for occasional transient reptiles to be present in these areas.

Birds

4.30 All of the boundary hedgerows and trees provide good quality potential nesting habitat for a range of species, including typical suburban bird species such as wren *Troglodytes troglodytes,* robin *Erithacus rubecula*, blackbird *Turdus merula* and dunnock *Prunella modularis*, as well as species generally associated with farmland, such as yellowhammer *Emberiza citronella* and linnet *Linaria cannabina*.

4.31 The grassland, prior to cutting, is likely to have also been suitable for ground nesting species such as red legged partridge *Alectoris rufa* and pheasant *Phasianus colchicus*, but is too small and surrounded by too many large hedgerows, trees and tall fencing to provide suitable nesting habitat for skylarks *Alauda arvensis.*

[redacted]

Water Vole and Otter

4.33 Records of water vole were returned from Holland Brook, over 1.5km to the north east of the site. The author is also aware of the presence of otter along this brook. Whilst the northern and eastern boundary ditch does ultimately connect to Holland Brook, this ditch remains essential to the drainage of the adjacent offsite arable field, and will be retained, unaffected and buffered as part of the proposals.

Dormice

4.34 The boundary hedgerows provide moderate quality potential habitat for dormice, however these habitats are not of a sufficient size to support a viable population of dormice, nor are they connected to any such habitats.

Invertebrates

4.35 The site is considered likely to support common and widespread invertebrate species typical o the habitats present. The moderately floristically rich grassland provides a valuable resource for foraging pollinators, likely to be of value at the site scale.

Other Legally Protected Species

4.36 Due to a lack of suitable habitats the site is not considered likely to support any other legally protected species.

Species of Principal Importance

4.37 The site provides potential foraging, nesting and shelter opportunities for Species of Principal Importance (SPIE) such as hedgehog *Erinaceus europaeus*, dunnock, yellowhammer, linnet, song thrush *Turdus philomelos*, starling *Sturnus vulgaris* and common toad *Bufo bufo*.

5.0 CONCLUSIONS AND RECOMMENDATIONS

Statutory Designated Sites

5.1 Holland on Sea Cliff SSSI, which is located c.2km to the east of the site, has no direct or indirect connectivity to the site, and no adverse impacts are predicted as result of the proposals. Whilst Holland Haven Marshes SSSI, located c.1km to the north east of the site, has no direct connectivity, it is noted that the northern and eastern boundary ditches form part of a tributary draining in the Holland Brook. The ditch had several running field drains at the time of survey, taking drainage water from the arable field offsite to the north. Local topography indicates that the development site does not drain into this ditch.

5.2 To ensure no fuel / chemicals can reach this ditch, a minimum 10m buffer strip of grassland will be retained between the ditch and the proposed construction zone, to be fenced off for the duratior of the construction works. Best practice measures relating to fuel / chemical storage will also be followed within the construction zone, to include appropriate spill kits and allocated re-fuelling zones.

5.3 The proposals are not considered likely to have any direct adverse impacts upon any national or international designated sites. However, the application site lies well within the recommended Zone of Influence for sites such as Hamford Water SAC, SPA and Ramsar site, the Colne Estuary (Mid Essex Coast Phase 2) SPA and Ramsar site, and the Essex Estuaries SAC with respect to recreational pressures (Colchester Borough Council, 2013). All internationally designated sites are fully protected by the Conservation of Habitats and Species Regulations 2017. Any new development must avoid having a significant adverse effect on the ecological features for which an SPA/SAC/Ramsar site was designated. Any such effect must be considered in combination with potential effects from other developments within influencing distance of the designated site.

5.4 Due to the local topography, existing habitats present on site, and the relatively small scale of the development, the proposals are very unlikely to have a direct adverse effect upon any of the above mentioned internationally designated sites, or any other such site within the region; however they are likely to contribute to cumulative impacts associated with increased visitor pressure.

5.5 A financial contribution to the emerging Essex Coast RAMS is likely to be required in order to ensure that there will be 'no likely significant effect' on the Hamford Water, Colne Estuary or Essex Estuaries designated sites. A per house tariff has been adopted and should be secured as part of the Essex Coast RAMS via the necessary means (Section 106 Agreement, Unil Undertaking etc).

5.6 The RAMS will work towards a range of locally appropriate and effective r ensure that increased local visitor numbers will not have an adverse impact upon any European designated site within the immediate Essex / south Suffolk region.

Non-Statutory Designated Sites

5.7 The proposals are not considered likely to be detrimental any LoWS. No further survey or mitigation is recommended.

Habitats

5.8 The boundary hedgerows are all Priority Habitats, listed under Section 41 of the Natural Environment and Rural Communities (NERC) Act 2006 (as amended). All are in 'good' condition, and all will be retained as part of the proposals.

5.9 No other Priority Habitats are present on site, and no further works with respect to habitats are necessary.

Amphibians

5.10 Great crested newts (GCNs) and their habitats are fully protected under the Conservation of Habitats and Species Regulations 2017 (as amended), and by the Wildlife and Countryside Act 1981 (as amended).

5.11 Potential effects: negligible.

5.12 Mitigation measures: none required.

5.13 Residual effects: none, due to negligible potential for GCN to be present locally. The loss of moderately species rich grassland will have a major adverse effect upon other spe amphibians at the site scale, however the retention of the hedgerows and grass buffer, plus th creation of flowering lawns, SUDS feature, small areas of scrub and a hibernacula will reduce the predicted impacts to minor adverse at the site scale.

Reptiles

5.14 All Essex reptile species are protected against harm under the Wildlife and Countryside Act 1981 (as amended).

5.15 Potential effects: negligible to minor adverse in the absence of mitigation measures.

5.16 Mitigation measures: there is moderate potential for occasional transient reptiles (common lizard and slow worm) to be present along the edges of the boundary hedgerows, and in small areas of longer, unmanaged vegetation primarily located along the south eastern site boundary.

5.17 The hedgerow margins will be retained. Works to remove vegetation along the south eastern boundary will be subject to a precautionary staged clearance using hand held strimmers. The vegetation will be strimmed to 150mm height, and left for 2 hours, before being strimmed to ground level. Vegetation will be cleared from the southern end of the site in a northerly direction to encourage any reptiles to move into offsite retained habitat. Such works will only take plac between mid-March and mid-October in dry, bright, warm (>10°C) weather to avoid harm to hibernating reptiles and ensure that reptiles are sufficiently warm to be able to move away.

5.18 **<u>Note:</u>** this approach is dependent upon the grassland being maintained in a regularly mown state, and not exceeding 150mm in length or providing dense vegetation cover, to prevent the wi site from becoming suitable for reptiles.

5.19 Residual effects: minor adverse to negligible, subject to the long term favourable management of the grassland buffer zones to provide areas of suitable reptile habitat, albeit at a smaller scale than that currently provided onsite. Margins of reptile habitat should be retained in these areas on a rotational basis to ensure that the floristic diversity of the meadow is enhanced, and t scrub does not encroach into the grassland from the boundary hedgerows. A hibernacula will provide new opportunities for hibernating reptiles onsite.

Birds

5.20 Breeding birds and their nests are protected under the Wildlife and Countryside Act 1981 amended).

5.21 Potential effects: the boundary hedgerows (to be retained) and to a lesser degree the open grassland provide suitable habitat for nesting birds, and the disturbance and destruction of an active nest could have a negative effect on some bird species at the site level.

5.22 Mitigation measures: vegetation clearance works should only be carried out during October to February inclusive to avoid the bird nesting season. If this is not possible, immediately prior commencement of works a check for nesting birds should be undertaken by experienced ecologist. Any active nests will need to be left in situ until the young have left the nest.

5.23 Residual effects: minor overall enhancement. The retention of the boundary hedgerows and the implementation of the measures detailed in section 6.0 will maintain the value of the site to nesting farmland bird species, and result in a significant enhancement at the site level for house sparrow, swift and starling. The retention of a boundary grass buffer and the creation of new areas flowering lawns, a SUDS feature, scrub and scattered trees will in part mitigate for the loss of grassland foraging habitat.

Bats

5.24 All species of bat are protected under the Conservation of Habitats and Species Regulations 2017 (as amended) and by the Wildlife and Countryside Act 1981 (as amended). In summary, th makes it an offence to harm or disturb a bat; damage or destroy a roost; and obstruct access t a roost (whether or not bats are present at the time).

5.25 Potential effects on roosting bats: negligible. All trees are due to be retained as part of the proposals, and none will be artificially illuminated at night.

5.26 Mitigation measures for roosting bats: none required, subject to the implementation of the lighting scheme detailed below.

5.27 Potential effects on commuting / foraging bats: with the exception of one multi-stemmed field maple, all of the boundary hedges and trees are to be retained, alongside a margin of existing grassland. The proposed dwellings are also set back from the site boundaries, and it is also noted that these boundary features have relatively poor connectivity to areas of offsite potential foraging / commuting / roosting habitat. It was not therefore considered necessary to carry out bat activity surveys. However, in the absence of mitigation the adverse effects of artificial illumination on some foraging and commuting bat species – particularly brown long-eared bats – could be greater where inappropriate lighting is installed.

5.28 Mitigation measures for commuting / foraging bats: lighting within the new development should be minimal, ideally limited to single porch lights only, located as close to the ground as possible and avoiding light spill towards the site boundaries. External lighting should be on short duration motion sensitive timers and use hoods, cowls, louvres and shields to direct light to the ground. Bulbs should be warm white (<3000K), and use LED bulbs of the lowest wattage possible.

5.29 Residual effects: following the provision of six artificial roosting features within the new development, a minor enhancement for crevice dwelling bat species is likely to result. Subject to the implementation of a wildlife friendly lighting scheme, the overall value of the site to foraging and commuting bats can be maintained with the retention of all boundary hedges and trees and an accompanying grass buffer, and the creation of a new SUDS feature, areas of flowering lawn mix and scattered native shrubs / trees.

Invertebrates

5.30 Potential effects: in the absence of mitigation, a major adverse impact is predicted at the site scale due to the loss of over 0.8ha of moderately floristically rich grassland.

5.31 Mitigation measures: the retention, enhancement and long term favourable managemen grassland adjoining the retained hedges / trees, and the creation of a new SUDS feature, grassland and scrub habitats of benefit to pollinators will mitigate for much of the loss of onsite habitats.

5.32 Residual effects: following implementation the proposed habitat enhancement measures detailed in section 6.0, a minor negative impact at the site scale on local invertebrate populations is predicted.

Dormice

5.33 Dormice and their habitats are fully protected under the Conservation of Habitats and Species Regulations 2017 (as amended) and by the Wildlife and Countryside Act 1981 (as amended).

5.34 Potential effects: negligible.

5.35 Mitigation measures: none.

5.36 Residual effects: negligible.

[redacted]

Otters & Water Voles

5.41 Otters and their habitats are fully protected under the Conservation of Habitats and Species Regulations 2017 (as amended) and by the Wildlife and Countryside Act 1981 (as amended). Water voles and their habitats are fully protected by the Wildlife and Countryside Act 1981 (as amended).

5.42 Potential effects: none.

5.43 Mitigation measures: none.

5.44 Residual effects: none.

Other Legally Protected or Notable Species

5.45 The proposed development is not anticipated to impact on any other legally protected species, therefore no mitigation measures are recommended.

5.46 The retention of existing habitat corridors and the creation of a SUDS feature, grass and scrub habitats will predominantly maintain the value of the site for species such as linnet, yellowhammer, toad, hedgehog, song thrush and dunnock. Onsite measures will increase the value of the site to nesting house sparrow, swift and starling (all SPIE); and to crevice dwelling roosting bats (many of which are SPIE).

6.0 MITIGATION & ENHANCEMENT MEASURES

6.1 **6 no. Habibat house sparrow triple terrace boxes** (or similar) will be built in to the northern or eastern façades of six garage dwellings (1 no. terrace per dwelling). The boxes will be located immediately beneath the eaves of the garages, at heights of c.3m and well away from all window and door openings. The recommended box type is shown below; others must be agreed with an ecologist.

Habibat House Sparrow Terrace Box

Made of concrete, to be integrated into buildings during construction. Can be sup with various brick facings, or without brick facings for incorporation into a ren weatherboarded wall.

440 x 215 x 150mm

Available from habibat.co.uk

6.2 **6 no. bat boxes** will be provided on site, ideally built into gable end walls at heights of 3-6m and facing between south east and south west. The boxes will be located immediately beneath the eaves or at the gable end apex, and well away from windows and sources of artificial lighti The boxes will have a 1-2m clear drop beneath the box entrance i.e. clear of all wires, aerials etc. The recommended box types are shown below; others must be agreed with an ecologist.

[redacted]

[redacted]

Bat Box

To fit in to the outside skin of 75mr course; or ***can be supplied without brick facii for incorporation into a weatherboarded wall***

Available from birdbrickhouses.co.uk

6.3 **<u>2 no. Habibat starling boxes</u>** will be built in to the northern or eastern façades of two dwellings (1 no. box per dwelling), at a height of 3m of more and ideally in a garage wall. Boxes should be located immediately beneath the eaves where possible and well away from window and door openings. The recommended box type is shown below; others must be agreed with an ecologist.

<u>Habibat Starling Box</u>

Made of concrete, to be integrated intc buildings during construction. Can be sup with various brick facings, or without bricl facings for incorporation into a ren weatherboarded wall. 215 x 215 x 120mm

Available from habibat.co.uk

6.4 **<u>9 no. swift nest boxes</u>** will be built in to the gable ends of two or three of the new buildings (groups of 3 no. boxes per building), with northern or eastern gables preferable but not essential for built-in boxes. The boxes will be positioned at least 4-6m high and must be placed at least 0.5m apart, ideally 1m, or spread across a gable end apex.

Only boxes shown on the webpages of Swift Conservation (https://www.swift-conservation.org/) should be used. A selection of these box types are shown below; others are available on the website.

<u>Build-in swift nest box</u>

The ideal internal depth of a swift box is 140mm, however if cavity width is limited, boxes ca be manufactured with a reduced depth (m 100mm). Can be supplied with various brick faci or without brick facings for incorporal rendered or weatherboarded wall.

Available from brickbirdhouses.co.uk

<u>Manthorpe swift brick</u>

Made of plastic, to be incorporated brick wall. Available in different colour fac

Available from wildcare.co.uk

<u>Swift S-brick</u>

To be incorporated into a brick wall. Avail in different colour facings, and for rer walls.

Available from actionforswifts.com

6.5 At least **1 no. reptile / amphibian hibernacula** measuring 1m x 2m x 1m will be created onsite, ideally hidden by developing shrubs close to the edge of the new SUDS feature. For longevity the hibernacula will consist primarily of rubble, with some logs, and with the top covered by soil which could also be seeded to speed colonisation by vegetation.

On-site Habitat Creation

6.6 All areas of **newly created grassland** outside of all private gardens, such as those around the proposed new SUDS feature, will comprise a **flowering lawn mix,** which is designed to provide valuable habitats for pollinators whilst also being relatively easily to maintain in the long term with basic management skills and tools, more aesthetically acceptable within a urban environmei and more realistically achievable within an urban environment.

6.7 The **new SUDS feature** will be designed to ensure part retains water for the majority of the year, and thereby functions more as a pond. The water body will be left to colonise naturally and not planted, however the margins and banks may be planted / sown with a marginal mix appropriate to the long term functionality of the water body.

6.8 All **tree and shrub planting** will be native, and appropriate for the location and long t management of the site e.g. blackthorn will not be planted as a scrub species, to avo potential for blackthorn suckers to spread across grassland in the future. Trees and shrubs will not be planted close to the southern banks of the SUDS feature, to prevent future issues with shading, but may be used on all other banks to provide valuable habitat, disguise features s hibernacula and to discourage public access to the water.

7.0 REFERENCES

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8.0 LEGISLATION

The Conservation of Habitats and Species Regulations 2017 (as amended)

8.1 The Conservation of Habitats and Species Regulations 2017 (as amen
safeguards for European Protected Sites and Species as listed in the Habitats Directive. As a result, the same provisions remain in place for European protected species, licensing requirements and protected areas after Brexit.

8.2 Species protected by the former European legislation includes great crested newt, all UK species, dormice and otter. A number of other plant and animal species are also included sucl as sand lizard, smooth snake and natterjack toad, however these additional species are rare, with restricted geographical ranges and specific habitat types.

8.3 Under The Conservation of Habitats and Species Regulations 2017 (as amended) it is an offence to:

Damage, destroy or obstruct access to an EPS breeding or resting place;

Deliberately capture, injure or kill an EPS (including their eggs);

Deliberately disturb an EPS, in particular any actions which may impair an animals ability to survive, breed or nurture their young; or their ability to hibernate or migrate; or which may significar affect the local distribution or abundance of the species to which they belong.

8.4 The legislation applies to all stages of amphibian life cycles (eggs, larvae and adult), and to active bat roosts even when they are not occupied at that particular time of year.

8.5 Natural England can, under certain circumstances, grant a licence to permit actions which would otherwise be unlawful, subject to the species concerned being maintained at a Favou Conservation Status and there being a true need for the proposed works to take place.

8.6 Special Protection Areas (SPAs) and Special Areas of Conservation (SACs) are also afford protection under the Conservation of Habitats and Species Regulations 2017 (as amended). Ramsar sites, which are designated under the Convention on Wet of International Importance (1971), are afforded the same level of protection as SPAs and SACs via national planning policy.

The Wildlife and Countryside Act 1981 (as amended)

8.7 The Wildlife and Countryside Act 1981 (as amended) provides varied
range of species including those already listed above.

8.8 Water vole are one of the species not listed under the Conservation of Habitats and Species Regulations 2017 (as amended), but are afforded the highest level of protection under the Wildlife and Countryside Act 1981 (as amended).

8.9 It is an offence to intentionally kill, injure or take a water vole, to intentionally or recklessly damage or destroy a structure or place used for shelter and/or protection, to disturb a water vole whilst occupying a structure and/or place used for shelter and protection, or to obstruct access to any structure and/or place used for shelter or protection.

8.10 Other species, such as common lizard, slow worm, adder and grass snake, are afforded less protection. For these species it is an offence to intentionally or recklessly kill or injure animals.

8.11 All active bird nests, eggs and young are protected against intentional destruction. Schedule 1 listed birds e.g. barn owls, kingfishers, are further protected from intentional and reckless disturbance whilst breeding.

8.12 Schedule 9 of The Wildlife and Countryside Act lists plant species for which it is an offence for a person to plant, or otherwise cause to grow in the wild. This includes Japanese Knotweed which, under the Environment Protection Act 1990 (as amended) is classed as 'controlled waste'. If any parts of the plant including stems, leaves and rhizomes are taken off-site they must be disposed of safely at a landfill site licensed to deal with such contaminated waste.

8.13 Sites of Species Scientific Interest (SSSI) are afforded protection by the Wildlife and Countryside Act 1981 (as amended).

[redacted]

The Protection of Mammals Act 1996 (as amended)

8.15 The Act protects all wild mammals against actions which have the intention of causing unnecessary suffering, including crushing and asphyxiation.

The Natural Environment and Rural Communities Act 2006 (as amended)

8.16 Under sections 40 and 41 of the Natural Environment and Rural Communities Act (NERC) 2006 local authorities have an obligation to have regard to the purpose of conserving biodiversity in carrying out their duties. The majority of UK legally protected species are listed under Section 41 the NERC Act.

8.17 Section 41 (S41) of the Natural Environment and Rural Communities (NERC) Act (2006) also requires the Secretary of State to publish a list of habitats and species which are of 'principal importance for the conservation of biodiversity' in England (Species of Principal Importance in England – SPIE). The S41 list is used to guide decision-makers, including local and regional authorities, in implementing their duty under Section 40 of the act to have regard to the conservation of biodiversity in England when carrying out their normal functions.

The Environment Act 2021 & National Planning Policy Framework (NPPF)

8.18 The Environment Act 2021 makes provision for biodiversity gain to be a condition of planning permission in England, with a minimum 10% BNG mandatory from January 2024. The 25 Year Environment Improvement Plan (DEFRA, 2023) sets out goals for improving the environment and leaving it in a better state for the next generation, and is supported by the National Planning Policy Framework (NPPF) (Ministry of Housing, Communities and Local Government 2024), which makes general provisions for the delivery of BNG.

8.19 The NPPF states that planning policies and decisions should contribute to and enhance the natural and local environment by *"minimising impacts on and providing net gains for biodiversity, including by establishing coherent ecological networks that are more resilient to current and future pressures and incorporating features which support priority or threatened species such as swifts, bats and hedgehogs".*

8.20 Plans should:

a) *" Identify, map and safeguard components of local wildlife-rich habitats and wider ecological networks, including the hierarchy of international, national and locally designated sites of importance for biodiversity; wildlife corridors and stepping stones that connect them; and areas identified by national and local partnerships for habitat management, enhancement, restoration or creation; and*

b) *promote the conservation, restoration and enhancement of priority habitats, ecological networks and the protection and recovery of priority species; and identify and pursue opportunities for securing measurable net gains for biodiversity."*

8.21 The NPPF also states that when determining planning applications, local planning authorities should consider that " *if significant harm to biodiversity resulting from a development cannot be avoided (through locating on an alternative site with less harmful impacts), adequately mitigated, or, as a last resort, compensated for, then planning permission should be refused*".

8.22 Locally specific polices set out what strategies need to be taken into account when delivering BNG, and may include Green Infrastructure Strategies and Local Nature Recovery Strategies in order that BNG may contribute to wider nature recovery plans.

Statutory Designated Sites

8.23 Under the National Parks and Access to the Countryside Act 1949 (as amended), stat conservation agencies were able to establish National Nature Reserves (NNRs), with provisions for these areas strengthened by the Wildlife and Countryside Act 1981 (as amended). They are managed to conserve their habitats or to provide special opportunities for scientific study of th habitats communities and species represented within them.

8.24 Local Nature Reserves (LNRs) can be declared by local authorities after consultation with t relevant statutory nature conservation agency under the National Parks and Access Countryside Act 1949 (as amended). LNRs are not subject to legal protection, but are afforded protection against damaging operations via byelaws, and against development planning policies.

Non-Statutory Designated Sites

8.25 Local Wildlife Sites (LWS), Sites of Importance for Nature Conservation (SINCs), Sites of Nature Conservation Importance (SNCIs) and County Wildlife Sites (CWS) are often designated by the local Wildlife Trust. They are not usually afforded ay legal protection, but are recognised in the planning system and given some protection through planning policy.

Appendix 1:

Proposed Site Layout

1. New vehicular access point with 90 metre vision splay shown
2. Minor Access combined pedestrian/vehicular access 5.8 metres wide.
3. Type 3 Turn
4. 550m^2 swale Suds feature
5. Blue dashed line shows pedestrian walking route around site perimeter - with active frontage address. Blue circles show link points to internal footpath infrastructure or to external footpaths and green spaces.
6. **29 dwellings shown**: mix of apartments; detached; semi-detached; 2, 3, 4 bedrooms; each with min. 100m^2 private amenity space, each with 2 no. parking spaces 2.9m x 5.5m, most with associated cartlodge or garage provision. Visitor parking spaces also shown.

0 10 20 30 40 50
metres

SCALE BAR 1:500

GENERAL NOTES

1. Dimensions are in millimetres (unless stated otherwise) and are to block or stud faces.
2. Drawings are not to be scaled, use figured dimensions only.
3. Notify the Architect of any discrepancies within the drawing and contact for clarification before proceeding.
4. All proprietary items to be fitted strictly in accordance with manufacturers instructions.
5. All works to be carried out in accordance with latest related British Standards and relevant codes of practice.

Rev	Date	Revision Description

Project Title: **PROPOSED RESIDENTIAL DEVELOPMENT, LAND OFF SLADBURY'S LANE**	Scale	1:500 @ A2
	Project Ref.	3735
	Drawing Ref.	PA-10
Drawing Title: **SITE PLAN PROPOSED**	Date Drawn	MAY 2025
	Drawn By	CR
	Checked By	CR

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RIBA Chartered Practice [redacted] CIAT Chartered Practice

OUTLINE PLANNING APPLICATION ISSUE

Appendix 2:

HSI Assessment Results

HSI Assessment results

Table 2: WB1

Habitat Suitability Index				
				SI value
SI1.	Map location	**A/B/C**	A	1.00
SI2.	Surface area	**rectangle/ellipse/irregular**	ellipse	
		length (m)	15	
		width (m)	12	
		OR estimate (m^2) if irregular		
		area (m^2) =	141.3	0.28
SI3.	Dessication rate	**never/rarely/sometimes/frequently**	rarely	1.00
SI4.	Water quality	**good/moderate/poor/bad**	good	1.00
SI5.	Shade	% of margin shaded 1m from bank	65	0.90
SI6.	Waterfowl	**absent/major/minor**	absent	1.00
SI7.	Fish population	**absent/possible/minor/major**	absent	1.00
SI8.	Pond density	number of ponds within 1km	3.8	1.00
SI9.	Terrestrial habitat	**good/moderate/poor/isolated**	poor	0.33
SI10.	Macrophyte cover	%	5	0.36
			HSI =	**0.70**
Use provisional HSI value if above 0.75			provisional HSI =	0.68
			Date undertaken	17.04.25

Table 3: WB2

Habitat Suitability Index				
				SI value
SI1.	Map location	**A/B/C**	A	1.00
SI2.	Surface area	**rectangle/ellipse/irregular**	ellipse	
		length (m)	65	
		width (m)	14	
		OR estimate (m^2) if irregular		
		area (m^2) =	714.35	1.00
SI3.	Dessication rate	**never/rarely/sometimes/frequently**	rarely	1.00
SI4.	Water quality	**good/moderate/poor/bad**	good	1.00
SI5.	Shade	% of margin shaded 1m from bank	0	1.00
SI6.	Waterfowl	**absent/major/minor**	absent	1.00
SI7.	Fish population	**absent/possible/minor/major**	absent	1.00
SI8.	Pond density	number of ponds within 1km	3.8	1.00
SI9.	Terrestrial habitat	**good/moderate/poor/isolated**	moderate	0.67
SI10.	Macrophyte cover	%	2	0.33
			HSI =	**0.86**
Use provisional HSI value if above 0.75			provisional HSI =	0.85
			Date undertaken	17.04.25

Appendix 3:

Great Crested Newt eDNA Results

Folio No:	771-2025
Purchase Order:	PO 1930
Contact:	Liz Lord
Issue Date:	13.05.2025
Received Date:	25.04.2025

SureScreen Scientifics

GCN eDNA Analysis

Summary

When great crested newts (GCN), *Triturus cristatus* , inhabit a pond, they continuously release small amounts of their DNA into the environment. By collecting and analyzing water samples, we can detect these small traces of environmental DNA (eDNA) to confirm GCN habitation or establish GCN absence.

Results

Lab ID	Site Name	OS Reference	Degradation Check	Inhibition Check	Result	Positive Replicates
CGN25 1103	Sladburys Lane - P1 North		Pass	Pass	Negative	0/12
GCN25 1104	Sladburys Lane - P2 South		Pass	Pass	Negative	0/12

Matters affecting result: none

Reported by: Amy Bermudez

Approved by: Chelsea Warner

SureScreen Scientifics

Methodology

The samples detailed above have been analyzed for the presence of GCN eDNA following the protocol stated in DEFRA WC1067 'Analytical and methodological development for improved surveillance of the Great Crested Newt, Appendix 5.' (Biggs et al. 2014). Each of the 6 sub-sample tubes are first centrifuged and pooled together into a single sample tube which then undergoes DNA extraction. The extracted sample is then analyzed using real-time PCR (qPCR), which uses species-specific molecular markers to amplify GCN DNA within a sample. These markers are unique to GCN DNA, meaning that there should be no detection of closely related species.

If GCN DNA is present, the DNA is amplified up to a detectable level, resulting in positive species detection. If GCN DNA is not present then amplification does not occur, and a negative result is recorded. Analysis of eDNA requires attention to detail to prevent the risk of contamination. True positive controls, negative controls, and spiked synthetic DNA are included in every analysis and these have to be correct before any result is declared and reported. Stages of the DNA analysis are also conducted in different buildings at our premises for added analytical security.

SureScreen Scientifics Ltd is ISO9001 accredited and participates in Natural England's proficiency testing scheme for GCN eDNA testing.

Interpretation of Results

Sample Integrity Check: When samples are received in the laboratory, they are inspected for any tube leakage, suitability of sample (not too much mud or weed etc.) and absence of any factors that could potentially lead to inconclusive results. Any samples which fail this test are rejected and eliminated before analysis.

Degradation Check: **Pass/Fail**. Analysis of the spiked DNA marker to see if there has been degradation of the kit or sample between the date it was made to the date of analysis. Degradation of the spiked DNA marker may lead indicate a risk of false negative results.

Inhibition Check: **Pass/Fail.** The presence of inhibitors within a sample is assessed using a DNA marker. If inhibition is detected, samples are purified and re-analyzed. Inhibitors cannot always be removed, if the inhibition check fails, the sample should be re-collected.

Result: **Presence of GCN eDNA (Positive/Negative/Inconclusive)**
Positive: GCN DNA was identified within the sample, indicative of GCN presence within the sampling location at the time the sample was taken or within the recent past at the sampling location.
Positive Replicates: Number of positive qPCR replicates out of a series of 12. If one or more of these are found to be positive the pond is declared positive for GCN presence. It may be assumed that small fractions of positive analyses suggest low level presence, but this cannot currently be used for population studies. In accordance with the WC1067 Natural England protocol, even a score of 1/12 is declared positive. 0/12 indicates negative GCN presence.
Negative: GCN eDNA was not detected or is below the threshold detection level and the test result should be considered as evidence of GCN absence, however, does not exclude the potential for GCN presence below the limit of detection.
Inconclusive: Controls indicate inhibition or degradation of the sample, resulting in the inability to provide conclusive evidence for GCN presence or absence.

Appendix 4:

Habitat Plan

KEY

- Other neutral grassland (g3c)
- Artificial unvegetated, unsealed surface (u1c)
- Developed land, sealed surface (u1b)
- Tall forbs (g3, 16)
- Individual trees
- Bramble scrub (h3d)
- Sparsely vegetated urban land - ruderal / ephemeral (u1f, 81)
- Native hedge with trees and ditch (h2a, 11, 50)
- Native hedge with ditch (h2a, 50)
- Site ownership boundary

(NB. Numbers relate to BNG habitat parcels)

TITLE:	Pre-development Habitat Plan
SITE:	Sladbury's Lane, Lt Clacton
CLIENT:	Stanfords
DATE:	03.04.25
DRAWN: Liz Lord	DWG NO. 1930_PreD
	SCALE: see drawing

Liz Lord Ecology